

13. What is the output of the following C program?

```
#include <stdio.h>
int update(int *p) {
    int x = *p;
    *(p + 1) += x;
    p++;
    *p += *(p + 1);
    return *p;
}
```

```
int main() {
    int a[4] = {2, 3, 4, 5};
    int r = update(a + 1);
    printf("%d %d %d %d %d", a[0], a[1], a[2], a[3], r);
    return 0;
}
```

- (A) 2 3 7 5 7
(B) 2 3 12 5 12
(C) 2 6 9 5 9
(D) 2 3 7 12 12
(E) 2 3 4 5 3

14. A system currently has two types of resources, R1 and R2. Each ordered pair (a, b) denotes a units of R1 and b units of R2. The available resource vector is Available = (2, 1). The Allocation and Max matrices for three processes are shown below:

	Allocation	Max
P0	(1, 0)	(3, 2)
P1	(2, 1)	(4, 2)
P2	(1, 1)	(2, 3)

(2, 1)
① → (4, 2)
② → ()

Which of the following is a valid safe sequence?

- (A) P0, P1, P2
(B) P2, P0, P1
(C) P0, P2, P1
(D) P1, P0, P2
(E) This state is unsafe, and no safe sequence exists